

Experiment/Case Study No : 9

Aim:

Implement localStorage and sessionStorage to Save User Preferences like Theme Settings.

Software/Tools Required

- Visual Studio Code
- Google Chrome (Web Browser)

Experiment Program Code

1. Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>localStorage and sessionStorage</title>
  <style>
    :root {
      --bg: #ffffff;
      --text: #1a1a1a;
      --card-bg: #f5f5f5;
      --accent: #4a90d9;
    }
    :root[data-theme="dark"] {
      --bg: #1a1a2e;
      --text: #e0e0e0;
      --card-bg: #16213e;
      --accent: #7c9ce0;
    }
    body {
      background: var(--bg);
      color: var(--text);
      font-family: system-ui, sans-serif;
      display: flex;
      justify-content: center;
      padding-top: 60px;
      transition: background 0.3s, color 0.3s;
    }
    .card {
      background: var(--card-bg);
      border-radius: 12px;
      padding: 32px;
      width: 340px;
      box-shadow: 0 4px 20px rgba(0,0,0,0.1);
    }
    h2 { margin-top: 0; }
```

```

label { display: block; margin: 16px 0 6px; font-weight: 600; }
select, input[type="range"] {
  width: 100%;
  padding: 8px;
  border-radius: 6px;
  border: 1px solid var(--accent);
  background: var(--bg);
  color: var(--text);
}
button {
  margin-top: 20px;
  width: 100%;
  padding: 10px;
  border: none;
  border-radius: 6px;
  background: var(--accent);
  color: #fff;
  font-size: 15px;
  cursor: pointer;
}
button:hover { opacity: 0.85; }
.row { display: flex; gap: 10px; }
.row button { flex: 1; }
#fontSizeValue { text-align: center; margin-top: 4px; font-size: 14px; }
</style>
</head>
<body>
<div class="card">
  <h2>Preferences</h2>

  <label for="theme">Theme</label>
  <select id="theme">
    <option value="light">Light</option>
    <option value="dark">Dark</option>
  </select>

  <label for="fontSize">Font Size: <span id="fontSizeValue">16px</span></label>
  <input type="range" id="fontSize" min="12" max="28" value="16" />

  <label for="language">Language</label>
  <select id="language">
    <option value="en">English</option>
    <option value="hi">हिन्दी</option>
    <option value="es">Español</option>
  </select>

  <div class="row">
    <button id="saveBtn">Save</button>
    <button id="resetBtn">Reset</button>
  </div>
</div>

<script>
const themeSelect = document.getElementById("theme");
const fontSizeInput = document.getElementById("fontSize");
const fontSizeValue = document.getElementById("fontSizeValue");
const languageSelect = document.getElementById("language");
const saveBtn = document.getElementById("saveBtn");
const resetBtn = document.getElementById("resetBtn");

```

```

function applyTheme(theme) {
  document.documentElement.setAttribute("data-theme", theme);
}

function applyFontSize(size) {
  document.body.style.fontSize = size + "px";
  fontSizeValue.textContent = size + "px";
}

function savePreferences() {
  const prefs = {
    theme: themeSelect.value,
    fontSize: fontSizeInput.value,
    language: languageSelect.value
  };
  localStorage.setItem("userPrefs", JSON.stringify(prefs));
  sessionStorage.setItem("userPrefs", JSON.stringify(prefs));
}

function loadPreferences() {
  const raw = localStorage.getItem("userPrefs") || sessionStorage.getItem("userPrefs");
  if (raw) {
    const prefs = JSON.parse(raw);
    themeSelect.value = prefs.theme || "light";
    fontSizeInput.value = prefs.fontSize || 16;
    languageSelect.value = prefs.language || "en";
    applyTheme(themeSelect.value);
    applyFontSize(fontSizeInput.value);
  } else {
    applyTheme("light");
    applyFontSize(16);
  }
}

function resetPreferences() {
  localStorage.removeItem("userPrefs");
  sessionStorage.removeItem("userPrefs");
  themeSelect.value = "light";
  fontSizeInput.value = 16;
  languageSelect.value = "en";
  applyTheme("light");
  applyFontSize(16);
}

saveBtn.addEventListener("click", savePreferences);
resetBtn.addEventListener("click", resetPreferences);

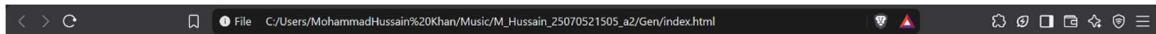
themeSelect.addEventListener("change", () => {
  applyTheme(themeSelect.value);
  savePreferences();
});

fontSizeInput.addEventListener("input", () => {
  applyFontSize(fontSizeInput.value);
});

fontSizeInput.addEventListener("change", savePreferences);
languageSelect.addEventListener("change", savePreferences);

```

```
window.addEventListener("storage", (e) => {  
  if (e.key === "userPrefs" && e.newValue) {  
    const prefs = JSON.parse(e.newValue);  
    themeSelect.value = prefs.theme;  
    fontSizeInput.value = prefs.fontSize;  
    languageSelect.value = prefs.language;  
    applyTheme(prefs.theme);  
    applyFontSize(prefs.fontSize);  
  }  
});  
  
loadPreferences();  
</script>  
</body>  
</html>  
Output
```

A 'Preferences' dialog box with a light gray background and rounded corners. It contains three settings: 'Theme' with a dropdown menu showing 'Light', 'Font Size' with a slider set to '17px', and 'Language' with a dropdown menu showing 'English'. At the bottom, there are two blue buttons labeled 'Save' and 'Reset'.

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Case Study Title

Design a task scheduler using local storage and session storage.

Case Study Program Code

1.index.html

```
<!DOCTYPE html>
<html lang="en" data-theme="light">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Seminar Schedule Planner</title>
  <style>
    :root {
      --bg: #f0f4f8;
      --text: #1a1a2e;
      --card-bg: #ffffff;
      --accent: #2c5f8a;
      --accent-hover: #1e4a6e;
      --border: #d1d9e6;
      --header-bg: #2c5f8a;
      --time-col: #e8f0f8;
      --stripe: #f5f8fb;
      --danger: #e74c3c;
      --danger-hover: #c0392b;
      --success: #27ae60;
      --warning: #f39c12;
    }
    :root[data-theme="dark"] {
      --bg: #0f1623;
      --text: #e0e6ed;
      --card-bg: #1a2332;
      --accent: #5a9fd4;
      --accent-hover: #3d7fb5;
      --border: #2d3a4e;
      --header-bg: #1e3a5f;
      --time-col: #162535;
      --stripe: #1e2d3f;
      --danger: #e74c3c;
      --danger-hover: #c0392b;
    }
  </style>

```

```

--success: #27ae60;
--warning: #f39c12;
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body {
  background: var(--bg);
  color: var(--text);
  font-family: 'Segoe UI', system-ui, sans-serif;
  font-size: 15px;
  padding: 20px;
  transition: background 0.3s, color 0.3s;
}
.container { max-width: 960px; margin: 0 auto; }
.top-bar {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 20px;
  flex-wrap: wrap;
  gap: 12px;
}
.top-bar h1 { font-size: 1.4rem; color: var(--accent); }
.prefs { display: flex; gap: 10px; align-items: center; }
.prefs select {
  padding: 6px 10px;
  border-radius: 6px;
  border: 1px solid var(--border);
  background: var(--card-bg);
  color: var(--text);
  cursor: pointer;
}
.card {
  background: var(--card-bg);
  border-radius: 12px;
  padding: 24px;
  box-shadow: 0 4px 16px rgba(0,0,0,0.08);
  margin-bottom: 20px;
}
.card h2 { margin-bottom: 16px; font-size: 1.15rem; color: var(--accent); }
.timetable {
  width: 100%;
  border-collapse: collapse;

```

```

    font-size: 14px;
    border-radius: 10px;
    overflow: hidden;
    border: 1px solid var(--border);
}
.timetable thead th {
    background: var(--header-bg);
    color: #fff;
    padding: 14px 12px;
    text-align: left;
    font-weight: 600;
    font-size: 13px;
    text-transform: uppercase;
    letter-spacing: 0.5px;
}
.timetable tbody td {
    padding: 12px;
    border-bottom: 1px solid var(--border);
    vertical-align: middle;
}
.timetable tbody tr:nth-child(even) { background: var(--stripe); }
.timetable tbody tr:hover { background: var(--time-col); }
.time-cell {
    background: var(--time-col);
    font-weight: 700;
    font-size: 13px;
    color: var(--accent);
    white-space: nowrap;
    min-width: 110px;
}
.status-badge {
    display: inline-block;
    padding: 4px 12px;
    border-radius: 20px;
    font-size: 12px;
    font-weight: 600;
}
.status-badge.completed { background: #d4edda; color: #155724; }
.status-badge.pending { background: #fff3cd; color: #856404; }
.status-badge.in-progress { background: #cfe2f3; color: #17a2b8; }
:root[data-theme="dark"] .status-badge.completed { background: #1e3a2a; color:
#7bc47f; }

```

```
:root[data-theme="dark"] .status-badge.pending { background: #3d3520; color:
#e0c060; }
:root[data-theme="dark"] .status-badge.in-progress { background: #1a2e4a; color:
#7ab3e0; }
.delete-btn {
padding: 6px 14px;
border: none;
border-radius: 6px;
background: var(--danger);
color: #fff;
font-size: 12px;
font-weight: 600;
cursor: pointer;
transition: background 0.2s;
}
.delete-btn:hover { background: var(--danger-hover); }
.add-btn {
margin-top: 16px;
padding: 10px 24px;
border: none;
border-radius: 6px;
background: var(--accent);
color: #fff;
font-size: 14px;
font-weight: 600;
cursor: pointer;
transition: background 0.2s;
}
.add-btn:hover { background: var(--accent-hover); }
.form-row {
display: grid;
grid-template-columns: 100px 1fr 1fr 120px auto;
gap: 10px;
margin-top: 16px;
align-items: end;
}
.form-row input, .form-row select {
padding: 8px 10px;
border-radius: 6px;
border: 1px solid var(--border);
background: var(--bg);
color: var(--text);
font-size: 14px;
```

```

}
.form-row .confirm-btn {
  padding: 8px 18px;
  border: none;
  border-radius: 6px;
  background: var(--success);
  color: #fff;
  font-size: 13px;
  font-weight: 600;
  cursor: pointer;
}
.form-row .confirm-btn:hover { opacity: 0.85; }
.alert-box {
  position: fixed;
  top: 50%;
  left: 50%;
  transform: translate(-50%, -50%) scale(0.8);
  background: var(--card-bg);
  border: 2px solid var(--accent);
  border-radius: 12px;
  padding: 28px;
  min-width: 300px;
  max-width: 400px;
  box-shadow: 0 10px 40px rgba(0,0,0,0.3);
  opacity: 0;
  pointer-events: none;
  transition: opacity 0.25s, transform 0.25s;
  z-index: 1000;
}
.alert-box.active { opacity: 1; pointer-events: auto; transform: translate(-50%, -50%) scale(1); }
.alert-box h3 { margin-bottom: 12px; color: var(--accent); font-size: 1.1rem; }
.alert-box p { line-height: 1.7; margin-bottom: 20px; font-size: 14px; }
.alert-box .btn-group { display: flex; gap: 10px; justify-content: center; }
.alert-box .btn-group button {
  padding: 8px 22px;
  border: none;
  border-radius: 6px;
  font-size: 13px;
  font-weight: 600;
  cursor: pointer;
}

```

```

.alert-box .btn-confirm { background: var(--danger); color: #fff; }
.alert-box .btn-confirm:hover { background: var(--danger-hover); }
.alert-box .btn-cancel { background: var(--border); color: var(--text); }
.alert-box .btn-cancel:hover { opacity: 0.8; }
.overlay {
  position: fixed;
  inset: 0;
  background: rgba(0,0,0,0.5);
  opacity: 0;
  pointer-events: none;
  transition: opacity 0.25s;
  z-index: 999;
}
.overlay.active { opacity: 1; pointer-events: auto; }
.day-header {
  background: var(--time-col);
  font-weight: 700;
  color: var(--accent);
  font-size: 13px;
  text-transform: uppercase;
  letter-spacing: 1px;
}
</style>
</head>
<body>
<div class="container">
  <div class="top-bar">
    <h1>Seminar Schedule Planner</h1>
    <div class="prefs">
      <select id="theme">
        <option value="light">Light</option>
        <option value="dark">Dark</option>
      </select>
    </div>
  </div>
  <div class="card">
    <h2>Seminar Timetable</h2>
    <table class="timetable" id="scheduleTable">
      <thead>
        <tr>
          <th>Time</th>

```

```

        <th>Session / Topic</th>
        <th>Speaker</th>
        <th>Room</th>
        <th>Status</th>
        <th>Action</th>
    </tr>
</thead>
<tbody id="scheduleBody"></tbody>
</table>
<button class="add-btn" id="addBtn">+ Add Session</button>
<div class="form-row" id="addForm" style="display:none;">
    <input type="time" id="fTime" />
    <input type="text" id="fTopic" placeholder="Topic" />
    <input type="text" id="fSpeaker" placeholder="Speaker" />
    <input type="text" id="fRoom" placeholder="Room" />
    <select id="fStatus">
        <option value="pending">Pending</option>
        <option value="in-progress">In Progress</option>
        <option value="completed">Completed</option>
    </select>
    <button class="confirm-btn" id="confirmAdd">Add</button>
</div>
</div>
</div>

<div class="overlay" id="overlay"></div>
<div class="alert-box" id="alertBox">
    <h3 id="alertTitle"></h3>
    <p id="alertMsg"></p>
    <div class="btn-group">
        <button class="btn-confirm" id="alertConfirm">Delete</button>
        <button class="btn-cancel" id="alertCancel">Cancel</button>
    </div>
</div>

<script>
const themeSelect = document.getElementById("theme");
const scheduleBody = document.getElementById("scheduleBody");
const addBtn = document.getElementById("addBtn");
const addForm = document.getElementById("addForm");
const confirmAdd = document.getElementById("confirmAdd");
const overlay = document.getElementById("overlay");

```

```

const alertBox = document.getElementById("alertBox");
const alertTitle = document.getElementById("alertTitle");
const alertMsg = document.getElementById("alertMsg");
const alertConfirm = document.getElementById("alertConfirm");
const alertCancel = document.getElementById("alertCancel");

const defaultSchedule = [
  { time: "09:00 - 09:30", topic: "Registration & Welcome", speaker: "Organizing Team", room: "Main Hall", status: "completed" },
  { time: "09:30 - 10:00", topic: "Inaugural Address", speaker: "Dr. A. Sharma", room: "Main Hall", status: "completed" },
  { time: "10:00 - 11:30", topic: "AI in Healthcare", speaker: "Prof. R. Mehta", room: "Main Hall", status: "in-progress" },
  { time: "11:30 - 12:00", topic: "Tea Break", speaker: "—", room: "Lobby", status: "pending" },
  { time: "12:00 - 13:30", topic: "Cloud Architecture Patterns", speaker: "Ms. P. Iyer", room: "Main Hall", status: "pending" },
  { time: "13:30 - 14:30", topic: "Lunch Break", speaker: "—", room: "Cafeteria", status: "pending" },
  { time: "14:30 - 16:00", topic: "DevOps & CI/CD Pipelines", speaker: "Mr. S. Kumar", room: "Main Hall", status: "pending" },
  { time: "16:00 - 16:30", topic: "Q&A & Vote of Thanks", speaker: "Panel", room: "Main Hall", status: "pending" }
];

let pendingDeleteIndex = -1;

function getSchedule() {
  const raw = localStorage.getItem("seminarSchedule") || sessionStorage.getItem("seminarSchedule");
  return raw ? JSON.parse(raw) : defaultSchedule;
}

function saveSchedule(data) {
  const json = JSON.stringify(data);
  localStorage.setItem("seminarSchedule", json);
  sessionStorage.setItem("seminarSchedule", json);
}

function getPrefs() {
  const raw = localStorage.getItem("userPrefs") || sessionStorage.getItem("userPrefs");
  return raw ? JSON.parse(raw) : { theme: "light" };
}

```

```

function savePrefs() {
  const prefs = { theme: themeSelect.value };
  localStorage.setItem("userPrefs", JSON.stringify(prefs));
  sessionStorage.setItem("userPrefs", JSON.stringify(prefs));
}

function applyPrefs(prefs) {
  themeSelect.value = prefs.theme || "light";
  document.documentElement.setAttribute("data-theme", themeSelect.value);
}

function renderTable() {
  const data = getSchedule();
  scheduleBody.innerHTML = "";
  data.forEach((item, idx) => {
    const tr = document.createElement("tr");
    tr.innerHTML = `
      <td class="time-cell">${item.time}</td>
      <td>${item.topic}</td>
      <td>${item.speaker}</td>
      <td>${item.room}</td>
      <td><span class="status-badge ${item.status}">${item.status.replace("-", "
    )}</span></td>
      <td><button class="delete-btn" data-idx="${idx}">Delete</button></td>
    `;
    scheduleBody.appendChild(tr);
  });
  scheduleBody.querySelectorAll(".delete-btn").forEach(btn => {
    btn.addEventListener("click", (e) => {
      e.stopPropagation();
      pendingDeleteIndex = parseInt(btn.dataset.idx);
      const item = getSchedule()[pendingDeleteIndex];
      alertTitle.textContent = "Confirm Delete";
      alertMsg.innerHTML = `Delete
<strong>${item.topic}</strong><br>${item.time} – ${item.speaker}?`;
      alertBox.classList.add("active");
      overlay.classList.add("active");
    });
  });
}

function hideAlert() {
  alertBox.classList.remove("active");
}

```

```

    overlay.classList.remove("active");
    pendingDeleteIndex = -1;
}

alertConfirm.addEventListener("click", () => {
    if (pendingDeleteIndex >= 0) {
        const data = getSchedule();
        data.splice(pendingDeleteIndex, 1);
        saveSchedule(data);
        renderTable();
    }
    hideAlert();
});

alertCancel.addEventListener("click", hideAlert);
overlay.addEventListener("click", hideAlert);

addBtn.addEventListener("click", () => {
    addForm.style.display = addForm.style.display === "none" ? "grid" : "none";
});

confirmAdd.addEventListener("click", () => {
    const time = document.getElementById("fTime").value;
    const topic = document.getElementById("fTopic").value.trim();
    const speaker = document.getElementById("fSpeaker").value.trim();
    const room = document.getElementById("fRoom").value.trim();
    const status = document.getElementById("fStatus").value;
    if (!time || !topic || !speaker || !room) {
        alertTitle.textContent = "Error";
        alertMsg.textContent = "Please fill all fields.";
        alertBox.classList.add("active");
        overlay.classList.add("active");
        return;
    }
    const data = getSchedule();
    data.push({ time, topic, speaker, room, status });
    saveSchedule(data);
    renderTable();
    document.getElementById("fTime").value = "";
    document.getElementById("fTopic").value = "";
    document.getElementById("fSpeaker").value = "";
    document.getElementById("fRoom").value = "";

```

```
document.getElementById("fStatus").value = "pending";
addForm.style.display = "none";
});

themeSelect.addEventListener("change", () => {
  document.documentElement.setAttribute("data-theme", themeSelect.value);
  savePrefs();
});

window.addEventListener("storage", (e) => {
  if (e.key === "seminarSchedule" && e.newValue) renderTable();
  if (e.key === "userPrefs" && e.newValue) applyPrefs(JSON.parse(e.newValue));
});

applyPrefs(getPrefs());
renderTable();
</script>
</body>
</html>
```

Output

Seminar Schedule Planner Light

Seminar Timetable

TIME	SESSION / TOPIC	SPEAKER	ROOM	STATUS	ACTION
09:00 - 09:30	Registration & Welcome	Organizing Team	Main Hall	completed	Delete
09:30 - 10:00	Inaugural Address	Dr. A. Sharma	Main Hall	completed	Delete
10:00 - 11:30	AI in Healthcare	Prof. R. Mehta	Main Hall	in progress	Delete
11:30 - 12:00	Tea Break	—	Lobby	pending	Delete
12:00 - 13:30	Cloud Architecture Patterns	Ms. P. Iyer	Main Hall	pending	Delete
13:30 - 14:30	Lunch Break	—	Cafeteria	pending	Delete
14:30 - 16:00	DevOps & CI/CD Pipelines	Mr. S. Kumar	Main Hall	pending	Delete
16:00 - 16:30	Q&A & Vote of Thanks	Panel	Main Hall	pending	Delete

[+ Add Session](#)

--:-- Topic Speaker Room Pending [Add](#)

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Result/Conclusion

The program was executed successfully, and the required output was obtained.

